

An on-chip micro-friction tester for tribology research of silicon based MEMS devices

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Abstract An on-chip micro-tribotester has been developed to investigate the friction and wear properties on side contacting surfaces of single crystal silicon that is most widely used in usual microelectromechanical systems actuators. The device is fabricated with standard bulk silicon process and bonding technology based on parameters that have been theoretically calculated to get the stiffness and friction forces. In this device, two comb shuttles are used. One comb shuttle is used to contact the friction surfaces under a certain normal load. The other comb shuttle moves back and forth to provide relative motion between the two friction surfaces. The tested two surfaces are the top surface of an anchor with rounded end and the lateral surface of a beam that has been connected to the two comb shuttles. Tribology experiments on the etched silicon surfaces that are side contacted have been carried out. Friction coefficients testing results suggest that dynamic friction coefficient is about 0.31–0.33 and the obtained static friction coefficient increases with the decrease of normal force. Wear experiment indicates that oxidation happens between the rubbing surfaces during the course of the testing. Wear debris is collected as agglomerates because of the function of adhesion force and water vapor and the agglomerates that are collected on top and lateral surfaces are of different shapes.

Keywords Friction · Wear · Dynamic friction coefficient · Static friction coefficient · Side contact · Silicon surface · Wear debris

1 Introduction

Microelectromechanical systems (MEMS) offer great promise for system integration of sensors, actuators and signal processing. The miniaturization and integration of MEMS devices are attractive in applications where smaller size and low weight are desirable. Despite the significant advances, there have always been major obstacles to the realization and reliability of many MEMS devices in past tribology problems. Research (Tadigadapa and Najafi 2001; Bushan 2001; Patton and Zabinski 2002; Komvopoulos 2003) indicates that with the dimensional decrease of MEMS devices, surface-to-volume ratios become larger and the size effects have great influence on the devices. Some of the forces such as frictional forces, which are pertinent to surface areas, become more and more important. This may greatly reduce the performance and life span of MEMS devices. Therefore, tribology issues in MEMS devices must be focused on to ensure reliable long-term operation in different environments.

Great progress has been made on micro-tribological research these years. Generally, there are two categories of testing systems to simulate friction and wear behaviors of MEMS devices, one is on-chip testing, and the other is off-chip testing. One of the off-chip experimental approaches typically uses atomic force microscopy (AFM) to evaluate tribology issues in which AFM tip is used to contact the surfaces. This approach is similar to the pin-on-plate method adopted in macroscale tribology measurements. Nevertheless, this kind of method cannot represent the real

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contact situation of most MEMS devices and the stiffness of the tip's cantilever may influence the results. Therefore, this kind of method cannot precisely simulate the actual friction and wear situation of MEMS devices. Another kind of off-chip experimental method uses a private tribology measurement system made by the researchers themselves (Blau and Budinski 1999; Schmidt et al. 2001; Chen and Carman 2000). This system can successfully represent the true contact situation of the surfaces. From many experiments they find that friction coefficients are no longer a constant at microscale. They are affected by many factors such as magnitude of normal force, contact area, difference of fabricating process and materials, etc. However, the researchers just focused on the top surface of MEMS materials, while another most popular contact situation, the lateral situation, has not been studied. The on-chip experimental system combines the contact pairs, driving system and force transducers on a single chip by MEMS technology. Compared with off-chip methods, on-chip testing systems have many advantages such as easy fabrication and realization of mass production. Many academic institutions such as Sandia National Laboratories (Miller et al. 1996; Walarven et al. 1999, 2003) and the University of California at Berkeley (Lim et al. 1990) are carrying out the tribology experiment by on-chip testing system. The object of their study is polysilicon. They discovered that the morphology of wear debris depended strongly upon the environment in which it is operated (such as humidity: low humidity causes particulate or rod-like debris, high humidity causes stringy debris).

This paper describes the design and fabrication of an on-chip bulk micromachined tribological test device. Its aim is to obtain fundamental information of the contact pair of the MEMS device. Fabricating the test structures in the same way as MEMS devices ensures that the material properties and chemical and morphological nature of the surfaces will be the same. Compared with the former on-chip research on polysilicon, this study focuses on single crystal silicon, which is also widely used for MEMS devices.

2 Structure and working principle

A simple figure of the tribotester is shown as Fig. 1. This system mainly consists of two comb-drive actuators, A and B, which are perpendicularly arranged in a plane. They are of the same figure and dimensions. Each shuttle has an extended beam in its moveable part, supported by three pairs of suspended beams laterally. The extended beams of the shuttles are perpendicularly connected in a plane. The contact pair of the system uses the flat-on-cylinder geometry with a 4 μm wide, 70 μm high rectangle cross-section beam rubbing against a rounded anchor as seen in Fig. 1b.

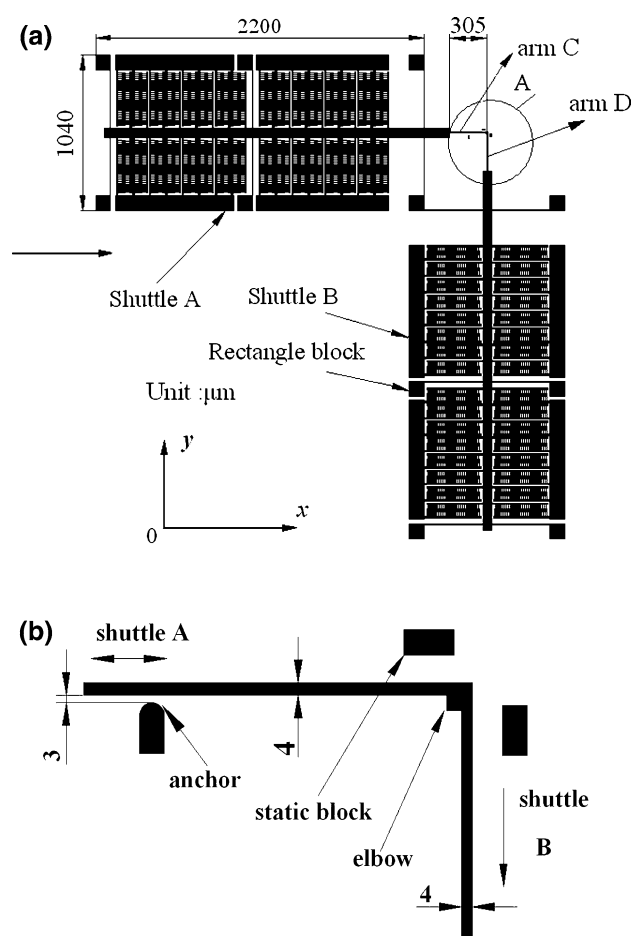


Fig. 1 Structure of the designed tribotester. **a** Configuration of the micro-tribotester. **b** Enlargement of portion A

Radius of the anchor's rounded end is 5 μm in and 70 μm in height. In uncharged state, there is a gap, 3 μm as designed, between the lateral surface of the beam and the rounded surface. The beam is brought into contact with the anchor by applying a DC voltage to comb drive B. Comb drive A reciprocates the beam back and forth against the anchor to initiate frictional force. In order to determine the position of the elbow, two perpendicularly arranged static blocks are placed on the edge of the system. Normal force and frictional force of the rubbing surfaces can be got by the relative displacement between the elbow and the stationary reference blocks.

3 Theoretical calculations

Theoretical calculations are carried out during the course of the design of the tribotester. Relationship between electrostatic force and driving voltage of the shuttles, stiffness of the tribotester together with frictional force and normal force calculations are all described.

3.1 Calculation of static frictional force

The tribotester uses two orthogonal comb-drive actuators to initiate normal force and frictional force, respectively. According to literature (Chen and Lee 2004), neglecting fringing effects, electrostatic force along the moving direction of the shuttle can be defined as

$$F_{\text{static}} = \frac{n\epsilon t}{g} v^2 \quad (1)$$

where n is the number of comb fingers, ϵ is the permittivity constant of air, t is the height of comb fingers, g is the gap between the comb fingers, v is driving voltage applied on the shuttle.

In a separate measurement when the beam is not in contact with the anchor, electrostatic force initiated by shuttle A must overcome the restoring force caused by the deflection of suspended beams of shuttle A and arm D in x direction. Their relationship can be described as

$$F_{\text{static}} = (k_{\text{beam}} + k_D)\delta_0 = k_x\delta_0 \quad (2)$$

where k_{beam} is stiffness of suspended beams of shuttle A. k_D is the stiffness of arm D in horizontal direction (x direction). k_x is the system's total stiffness in horizontal direction (x direction), $k_x = k_{\text{beam}} + k_D$. δ_0 is relative displacement of shuttle A in x direction.

Figure 2 is a simple calculating model of a single comb-drive actuator. According to the theory on mechanics of materials, k_{beam} can be expressed as

$$k_{\text{beam}} = \frac{72EI}{l^3} \quad (3)$$

where l is the length of the suspended beams, E is elastic modulus of single crystal silicon, I is moment of inertia of a single suspended beam, $I = \frac{b^3h}{12}$, b and h refer to the width and height of the suspended beams in the bending direction.

k_D in equation (2) can be calculated on the assumption that the root of the extended arm of shuttle B is fixed and the root of shuttle A can just move along horizontal

direction. Simplified computing model is shown as Fig. 3. According to elastic beam theory, k_D can be expressed as

$$k_D = \frac{F_1}{\delta_h} = \frac{12EI(l_1 + 3l_2)}{(4l_1 + 3l_2)l_2^3} \quad (4)$$

where l_1 and l_2 are lengths of the extended beams of shuttle A and shuttle B.

When a certain DC driving voltage is applied on shuttle B and the frictional pair is brought into contact, F_{static} exerted by shuttle A must overcome not only restoring force caused by the deflection of suspended beams of shuttle A and arm D, but also frictional force f between the contact pair. Corresponding to Eq. (2), their relationship can be described as

$$F_{\text{static}} = f + k_x\delta' \quad (5)$$

where δ' refers to relative displacement between the rubbing surfaces when the frictional pair is brought into contact.

From (2) and (5), frictional force f can be deduced as

$$f = k_x(\delta_0 - \delta') \quad (6)$$

3.2 Critical driving voltage calculation

In the design of the tribotester, shuttle B is used to initiate normal force for the frictional pair. Critical driving voltage, which decides when contact will happen, is one of the important parameters for the actuator. Because there is a $3\text{ }\mu\text{m}$ initial gap between the rubbing surfaces, the lateral surface of the extended beam of shuttle A must move at least $3\text{ }\mu\text{m}$ toward the rounded surface of the anchor before force balance is achieved. According to Eqs. (1) and (3),

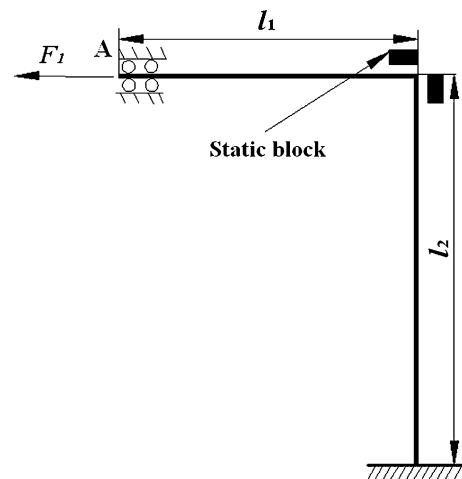


Fig. 3 Simplified calculating model on horizontal direction

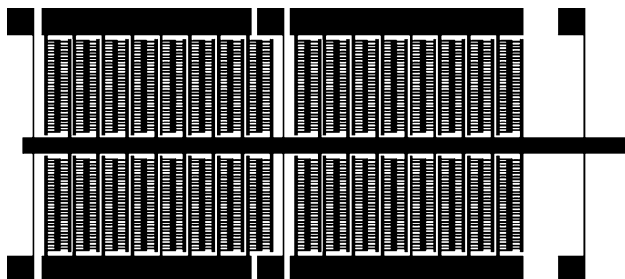


Fig. 2 Structure of single comb drive actuator

critical driving voltage, which is applied on shuttle B to initiate contact between the contact pair, can be expressed as

$$v_{\text{critical}} = \sqrt{\frac{3(k_c + k_{\text{beam}})g}{n\epsilon t}} = \sqrt{\frac{3k_y g}{n\epsilon t}} \quad (7)$$

where k_c refers to the stiffness of arm C in vertical direction, k_y is the system's total stiffness in vertical direction (y direction), $k_y = k_{\text{beam}} + k_c$. Simple calculating model of k_c is shown as Fig. 4. This model is built on the assumption that the root of the extended arm of shuttle A is immovable and the root of the extended arm of shuttle B can just move along vertical direction. According to elastic beam theory, k_c can be expressed as

$$k_c = \frac{12EI(l_2 + 3l_1)}{(4l_2 + 3l_1)l_1^3} \quad (8)$$

When the driving voltage applied on shuttle B is larger than v_{critical} , shuttle B will then go on moving in vertical direction. Comparing to the position caused by the applied critical voltage, there is a relative displacement δ_R at the elbow. Corresponding relationship between normal force N and relative displacement δ_R can be expressed as

$$N = \gamma \delta_R \quad (9)$$

where γ is a coefficient that is pertinent to the dimensions of the tribotester.

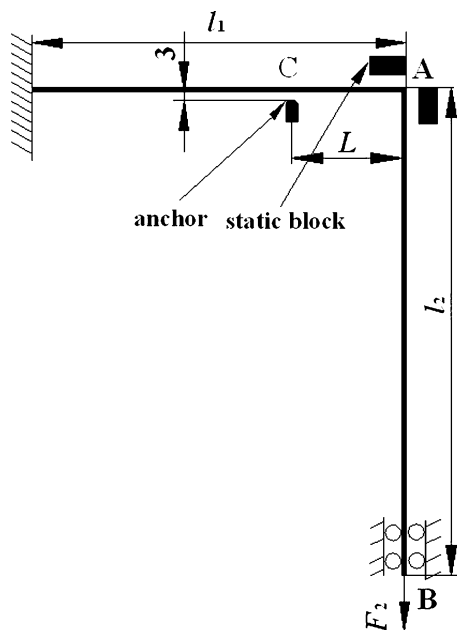


Fig. 4 Simplified calculating model on vertical direction

4 Fabrication

The tribotester is fabricated by standard MEMS bulk silicon technology and bonding process in which single crystal silicon is used as the material of shuttles and glass is used as substrate. Simple fabrication process is shown as Fig. 5. The process is started with a 4 in. single crystal silicon wafer which serves as raw materials of the tribotester (Fig. 5a). Then, the oxidated wafer is put into 40% KOH solution to create shallow groove by wet etching process. The groove (Fig. 5b), which is used to prepare for the creation of suspending part of the shuttles, is 4 μm in depth. Then, boron is infiltrated into silicon wafer to increase the conductivity of the tribotester (Fig. 5c). Subsequently, a piece of glass is picked to serve as substrate of the tribotester. On this substrate, a layer of Cu (400 $\text{\AA}$ thick), which serves as seed layer of the electrode, is deposited by sputtering technology (Fig. 5e). By lift-off technology, unwanted portion of the Cu film is gotten rid of and electrodes are left (Fig. 5f). Then, the infiltrated silicon wafer and glass are combined by bonding technology (Fig. 5g). Because practical thickness of the silicon wafer is much bigger than the designed value, the bonded silicon wafer is thinned to 75 μm by 40% KOH solution (Fig. 5h). Finally, inductively coupled plasma (ICP) technology is used to create comb-drive actuators and the testing part

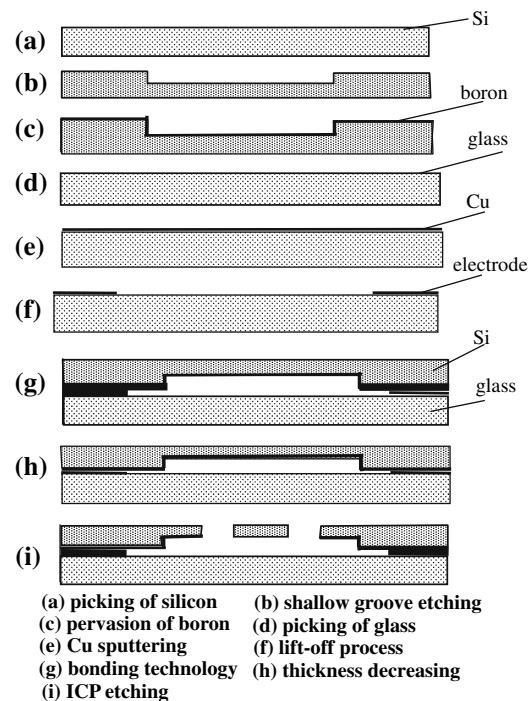


Fig. 5 Fabrication technology a picking of silicon b shallow groove etching c pervasion of boron d picking of glass e Cu sputtering f lift-off process g bonding technology h thickness decreasing i ICP etching

(as shown in Fig. 5i). Scanning electron microscope (SEM) photograph of the created structure is shown as Fig. 6.

5 Experiment and discussion

Morphology on top and lateral surfaces is tested with AFM when the system is fabricated with MEMS technology. The tested result is shown as Fig. 7. It is concluded from the figure that the roughness of the top and lateral surface are 104 and 35 nm, respectively, which implies that the morphology on the top surface is obviously different from that on the lateral surface and it is necessary to perform the tribology research on the lateral surface. Before the tribology experiment, the tribotester's stiffness k_x and k_y in x and y directions are also calibrated by probe station in order to increase the precision of the tested tribology result. The calibrating device is shown as Fig. 8. In this sketch the probe is fixed on a three dimensional platform that can move along x , y and z direction. The connection between the root of the probe and the workbench is considered to be rigid. The stage is firstly manipulated to ensure the tip of the probe touch with the side end of the chip. Then, the platform is forced to move a distance x_1 along the x direction. Meanwhile, the corresponding displacement of the end point of the chip, x_2 , is measured under the digital microscope. Then, the corresponding deformation of the tip of the probe is $x_1 - x_2$. According to Newton's third law, the restoring force caused by the deflection of the probe is equal to the force caused by the deformation of the structure to be calibrated. This can be expressed as

$$k_{\text{probe}}(x_1 - x_2) = k_x x_2 \quad (10)$$

where k_{probe} is the stiffness of the probe in x direction, and has been measured beforehand.

Thus

$$k_x = \frac{k_{\text{probe}}(x_1 - x_2)}{x_2} \quad (11)$$

In the same way, the stiffness k_y in y direction can also be calibrated. The calibrated values of k_x and k_y are 89 and 95 $\mu\text{N}/\mu\text{m}$, which are close to the theoretically calculated results 84 and 91 $\mu\text{N}/\mu\text{m}$.

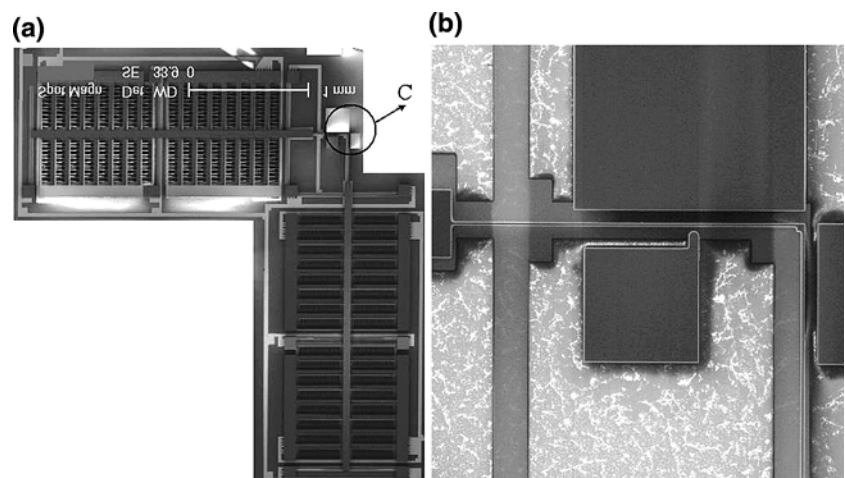
Static friction coefficient, dynamic friction coefficient and wear experiments are then tested in clean booth (1,000 class, 23°C room temperature and 21% atmospheric humidity) after the calibration of the tribotester's stiffness. Simple experiment theory is shown as Fig. 9. The testing system is mainly composed of power supply system, processing circuit, microscope, CCD image acquisition system (the rectified dimension of every pixel is 0.1 μm , which is precise enough to carry out the experiment) and computer. Power supply system in this experiment offers two sets of driving voltages. One is applied on shuttle B to produce normal force and the other is operated on shuttle A to initiate friction. Working condition of the rubbing surface is enlarged by microscope and captured by CCD. Then CCD transports the image data to computer by AD/DA datum collecting card. Computer recovers the images and displays them on the screen of the computer. By special image processing software, frictional forces and normal forces can be acquired by the deflections of the elbow. Static or dynamic friction coefficients are gained correspondingly.

5.1 Static friction coefficient testing

5.1.1 Experimental results

Static friction coefficient testing is carried out on the tribotester. During the experiment a series of ramped DC

Fig. 6 SEM sketch of the micro-tribotester **a** structure of tribotester **b** enlarged portion of C



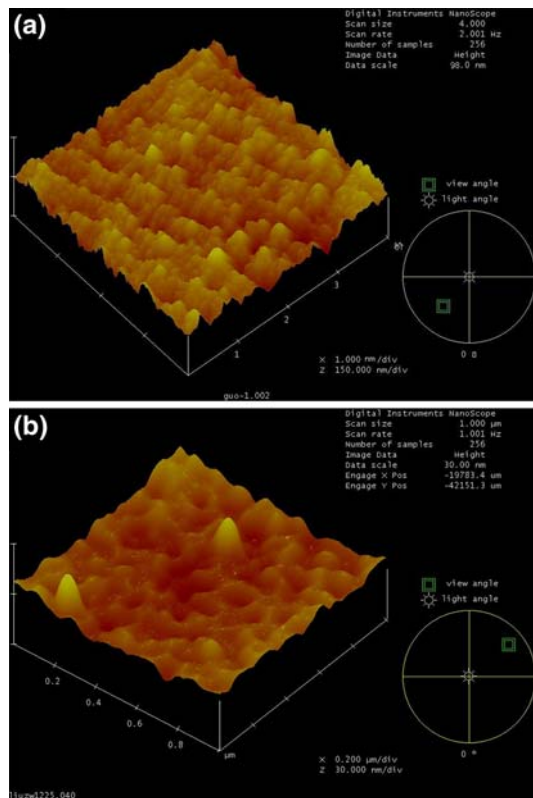


Fig. 7 AFM sketch of the roughness of the tribotester. **a** Lateral surface. **b** Top surface

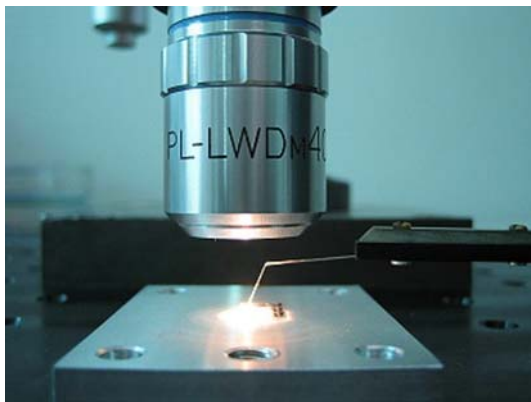


Fig. 8 Stiffness calibrating sketch of the tribotester

driving voltage v_A (0–30 V, 1 V increment) is firstly applied on shuttle A when the frictional surfaces are not in contact. Corresponding relative displacements δ_0 of the elbow are recorded. Then, v_A is got rid of and a certain DC v_B ($v_B > v_{\text{critical}}$) is applied on shuttle B to make the frictional pair contact. Similar ramped driving voltages v_A from 0 to 30 V is then applied on shuttle A to initiate relative displacement between the rubbing surfaces. Corresponding relative displacements δ' are also recorded.

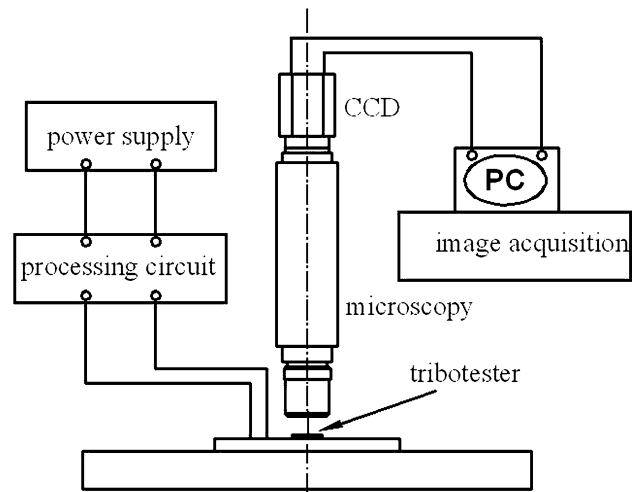


Fig. 9 Experimental sketch

Friction forces can be calculated by Eq. (6). Figure 10 shows the relationship between the rubbing surfaces' relative movement and driving voltages v_A when a certain DC driving voltage is applied on shuttle B ($v_B > v_{\text{critical}}$). The figure shows that when v_A is gradually increased at first, no relative movement happens between the frictional pair because of static friction force. But when v_A is larger than critical driving voltage V_{cri} , at which electrostatic force is equal to the biggest static frictional force, slip happens between the rubbing surfaces. Go on increasing v_A when it exceeds critical driving voltage, no stick takes place then. Table 1 lists the corresponding tested results of V_{cri} when 37, 40, 45 V driving voltage are applied on shuttle B, respectively. It is verified from the table that with the increment of v_B , V_{cri} increased correspondingly. Figure 11 depicts the relationship between μ (the ratio of frictional

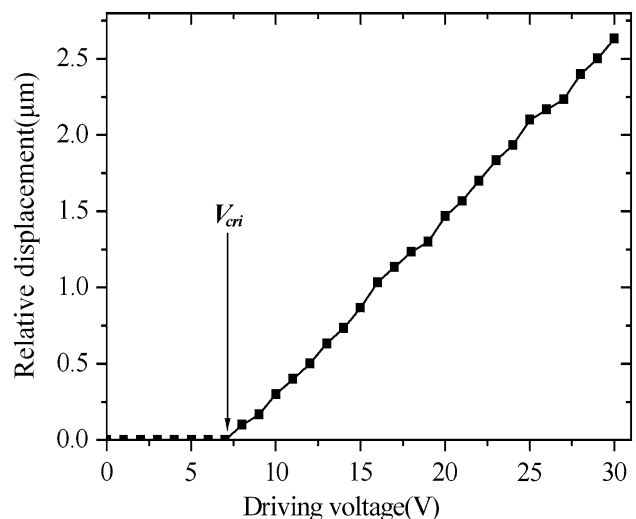
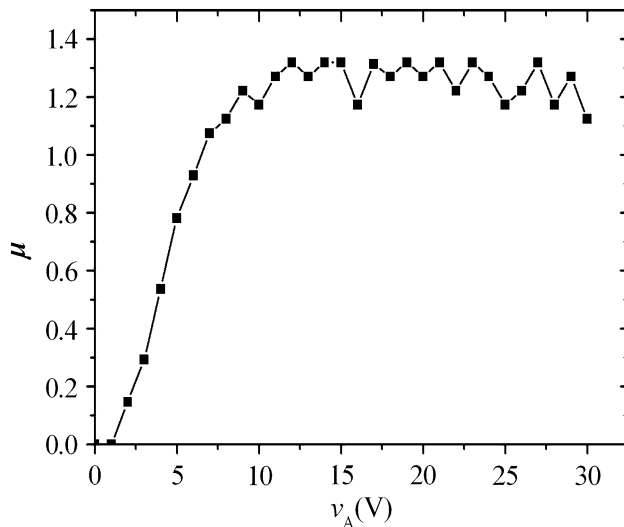


Fig. 10 Relationship between V_A and relative displacement

Table 1 Tested results of V_{cri}

Driving voltage v_B (V)	V_{cri} applied on shuttle A (V)
37	5
40	7
45	10


Fig. 11 Relationship between v_A and μ

force to normal force) and v_A . It is concluded that with the increment of v_A , μ increases with it at first. When v_A exceeds the critical value V_{cri} , relative slip happened between the contact pair. Go on increasing the driving voltage, relative slip occurs with it and μ is prone to be constant. This constant value can be considered as static friction coefficient. General scope and the average value of the static friction coefficient corresponding to 37, 40 and 45 V DC voltage are shown as Table 2. For each test condition, the experiment was repeated six times to increase the credibility of the tested data. The data in Table 2 is the average value over six measurements. It indicates in the table that with the increment of v_B (which means normal force increased correspondingly), static friction coefficient is no longer a constant as conventional frictional law. It may decrease with the increase of normal force.

5.1.2 Discussion of results

According to binomial theorem of tribology, frictional force is the synthetical function of physical (such as van der Waals force, etc.) and mechanical effect. The total frictional force is often expressed as

Table 2 Data of static friction coefficient

Driving voltage (V)	Corresponding normal force (μN)	Static frictional coefficient scope	Average value
37	25	1.37–1.68	1.526
40	39	1.12–1.32	1.27
45	62	0.95–1.1	1.02

$$F = \alpha A + BW \quad (12)$$

where

αA is the physical effect that constitute the frictional force, α is positive constant closed to 1, A is the real contact area between the contact pair, BW is the mechanical effect that constitute the frictional force, β represents the constant coefficients determined by the mechanical effects, W is the normal force.

In macroscale condition, because the contribution of surface effect to frictional force is negligible comparing to the mechanical effect (i.e., $\alpha A \ll \beta W$), under this condition Eq. (12) can be approximatively expressed as

$$F = \beta W \quad (13)$$

Equation (13) is the conventional Amontons law, where β is just the friction coefficient between the contact pair.

However, when the contacting dimension of the frictional pair is reduced to micrometer order, because of the “scale effect” under micro circumstance, surface effect becomes more dominant than macro conditions. It occupies much larger proportion in the composition of the frictional force. According to the definition of the friction coefficient, f can be expressed as

$$f = \frac{F}{W} = \frac{A}{W} \alpha + \beta \quad (14)$$

According to binomial theorem of tribology, when the deformation between the contact is elastic, the real contact area is proportional to $W^{2/3}$. Under this condition Eq. (14) can be substituted as

$$f = \lambda W^{-\frac{1}{3}} + \beta \quad (15)$$

Where λ is a positive constant that is pertinent to the contact status of the contact pair.

According to Eq. (15), when W increased, f decreased with it. Because single crystal silicon is a kind of elastic material that almost has no plastic deformation, we can consider the deformation between the contact pair is elastic. So with the increment of normal force, friction coefficient decreases with it.

5.2 Dynamic friction coefficient testing

Dynamic friction coefficient testing is also performed on this system when a series of biased sine potentials are applied on shuttle A. Table 2 shows the tested dynamic friction coefficient values on condition that 40, 45 V DC driving voltage are applied on shuttle B. Average values of the result are 0.31 and 0.33. They are almost equal to each other. This shows that normal forces may have less effect on dynamic friction coefficient than on static friction coefficient. It also indicates that dynamic coefficient is much less than static friction coefficient (Table 3).

5.3 Wear testing

Wear experiment is also carried out on the tester. Simple sketch of the testing structure is shown as Fig. 12. During the course of the experiment the DC driving voltage applied on shuttle B is 45 V. Corresponding normal force between the rubbing pair is $62 \mu\text{N}$. The biased sine driving voltage applied on shuttle A, which is used to initiate reciprocation between the rubbing pair, is $14 + 14 \times \sin(2 \times \pi \times 3000 \times t)$ V. Corresponding amplitude of the relative displacement of the contact pair is about $10 \mu\text{m}$. Element constitute of the contact pair is also calibrated by SEM before the experiment (shown as Fig. 13). From the figure we conclude that the main element of the contact pair is single crystal silicon (element constitute is 97% or so). Carbon (nearly zero) and oxygen (nearly 3%) are so little that they can be neglected. Figure 14 is the element constitute of the wear debris when the contact pair worked 55 h later. In these 55 h the contact pair has worked 574,200,000 cycles and the corresponding relative displacement is 5,742 m. Comparing with Fig. 13, oxygen is added from 3 to 21%. This indicates that oxidation reaction has happened between the rubbing surfaces. This may be caused by high temperature because of high frequency relative friction. This also indicates that chemical reaction sometimes has great effect on the result of

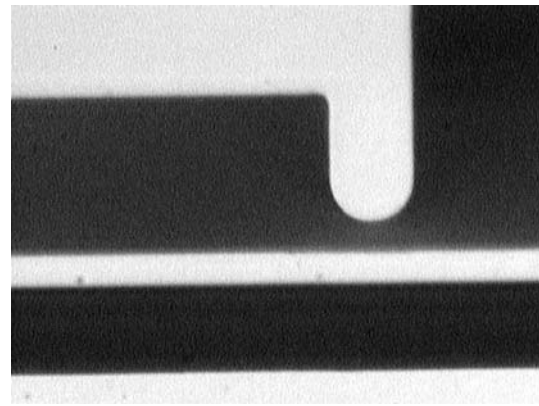


Fig. 12 Sketch of the contact pair

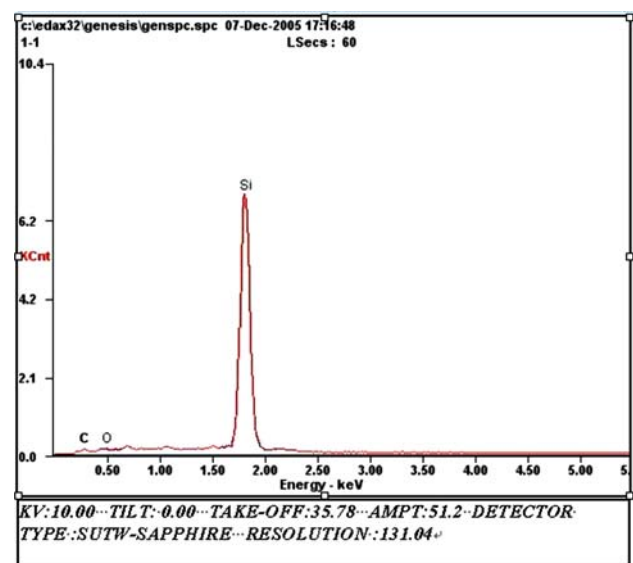


Fig. 13 Element constitute of the rubbing surfaces before wear experiment

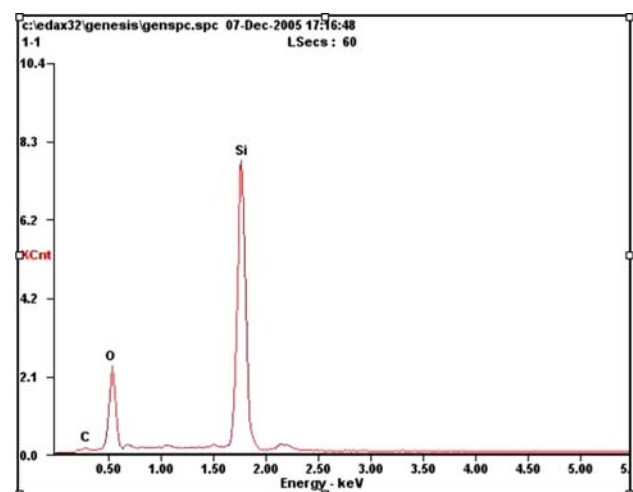
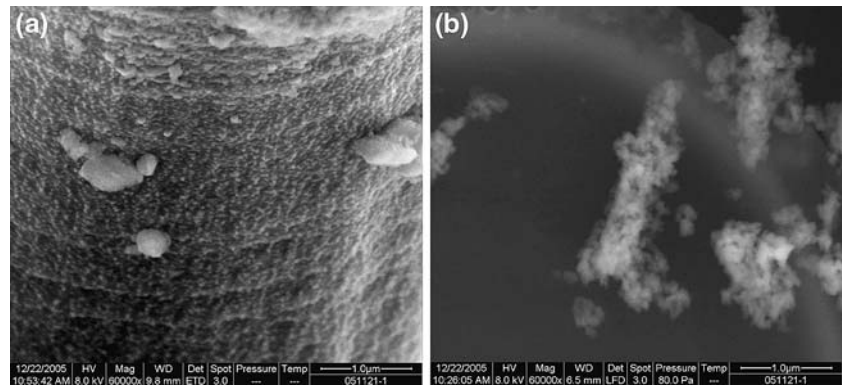


Fig. 14 Element constitute 55 h later

Table 3 Data of dynamic friction coefficient

Driving voltage (V)	Normal force (μN)	Dynamic friction coefficient	
		Measured value	Average value
45	62	0.36542	0.33
		0.29234	
		0.3222	
40	39	0.28374	0.31
		0.34823	
		0.31598	

Fig. 15 Comparison of the wear debris on lateral and top surfaces **a** debris on lateral surface **b** debris on top surface



experiment. It is unilateral just consider the effect of the physical factor when the tribology experiment is analyzed.

Figure 15 is the SEM sketch of the debris on lateral surface and top surface. The figure indicates that the wear debris is gathered as agglomerates but not the silica particles as usual conditions. According to (Smallwood et al. 2007), this may be formed because of the function of water vapor in the air and the great adhesion force between the silica particles because of the “scale effect”. It is also verified from the figure that the shape of the agglomerates on lateral surface are different from those on top surface.

6 Conclusions

An on-chip micro-tribotester has been successfully designed, fabricated and operated to simulate the tribology properties of single crystal silicon MEMS devices. The tester completely integrates two comb-drive actuators, the contact pair, and force sensors on a single chip by standard MEMS technology. Theoretical and experimental results indicate the following conclusions:

1. The tribotester can successfully simulate the friction and wear conditions on side contacting surfaces of single crystal silicon MEMS material.
2. The static friction coefficient is tested under microscope while the system is being fabricated. Experimental results indicate that normal force may have great influence on the static friction coefficient at microscale. When normal force increases, the static friction coefficient is no longer a constant value at microscale. It may decrease with the increment of normal force.
3. Dynamic friction coefficients of the frictional pair are 0.33 and 0.31 when 40 and 45 V DC of driving voltage are applied on shuttle B. They are almost of the same values.

4. The wear experiment indicates that oxidation happens between the rubbing surfaces during the course of the testing. Wear debris is collected as agglomerates because of the function of adhesion force and water vapor. The collected debris that agglomerates on top and lateral surfaces are of different shapes.

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